IAS PRELIMS/MAINS

SCIENCE & TECHNOLOGY

A COMPREHENSIVE COVERAGE

MODULE – 44

NANO TECHNOLOGY - 2

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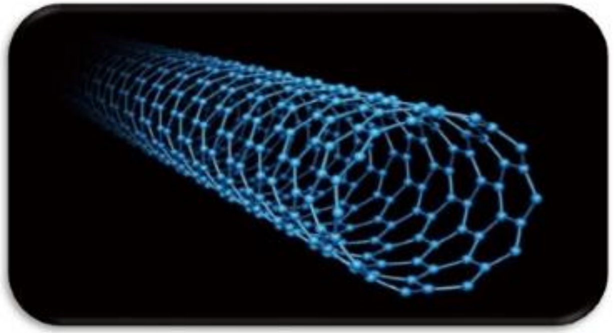
NANOMATERIALS

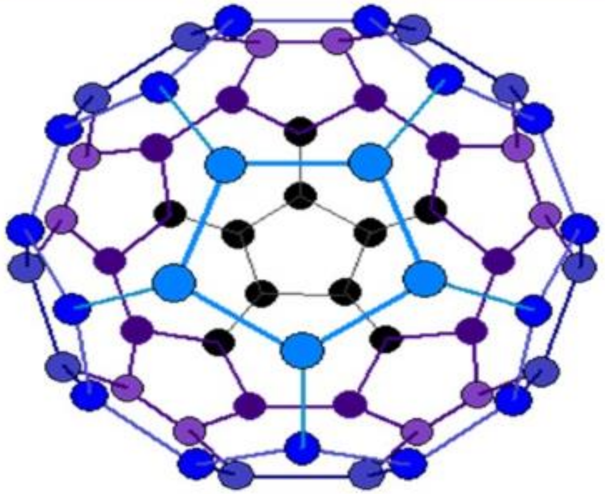
2 CARBON NANOTUBE

- Carbon nanotubes are cylindrical molecules that consist of rolled-up sheets of single-layer carbon atoms (graphene).
- They can be single-walled with a diameter of less than 1 nanometer (nm) or multi-walled with diameter reaching more than 100 nm.
- Their length can reach several micrometers or even millimeters.



3 BUCKMINSTERFULLERENE (C60)

- The size of the molecule is almost exactly 1 nm in diameter.
- The ratio of the size of an ordinary soccer ball to the planet Earth is the same as the ratio of the size of a C_{60} molecule to a soccer ball.



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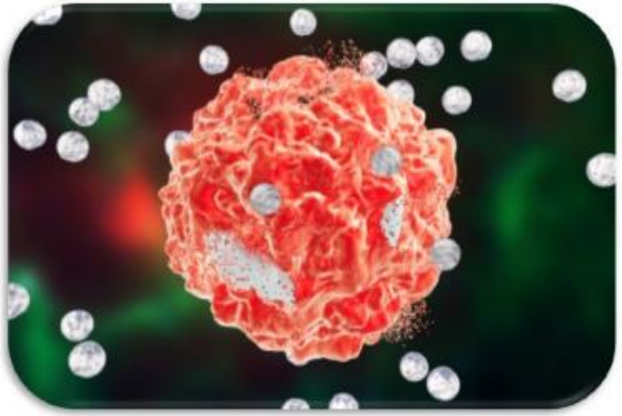
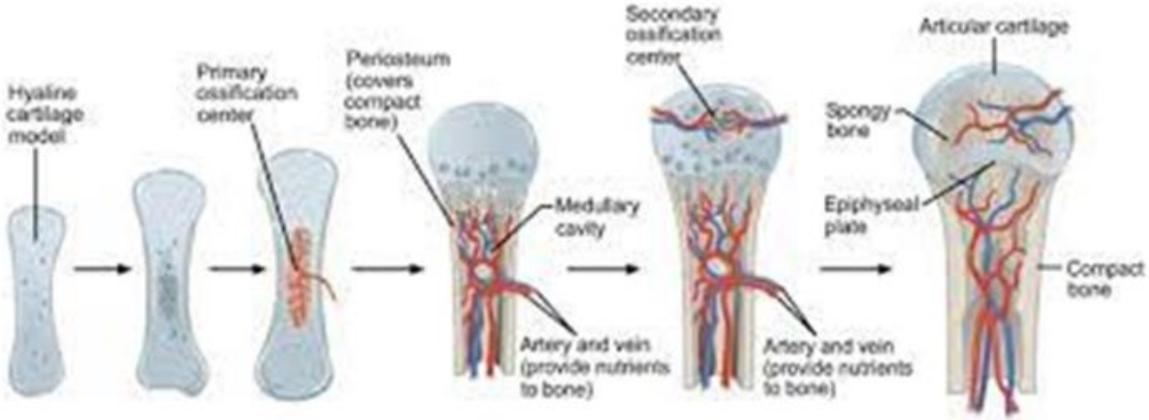
OTHER APPLICATIONS

- **Better imaging and diagnostic tools enabled by nanotechnology are paving the way for earlier diagnosis, and better treatment success rates.**
- Research in the use of nanotechnology for **regenerative medicine** - Ex: Materials can be engineered to mimic the mineral structure of human bone or materials can be made for dental applications.
- **Researchers** are looking for ways to **grow complex tissues with the goal of one day growing human organs for transplant.** Researchers are also studying ways to use graphene to help repair spinal cord injuries.





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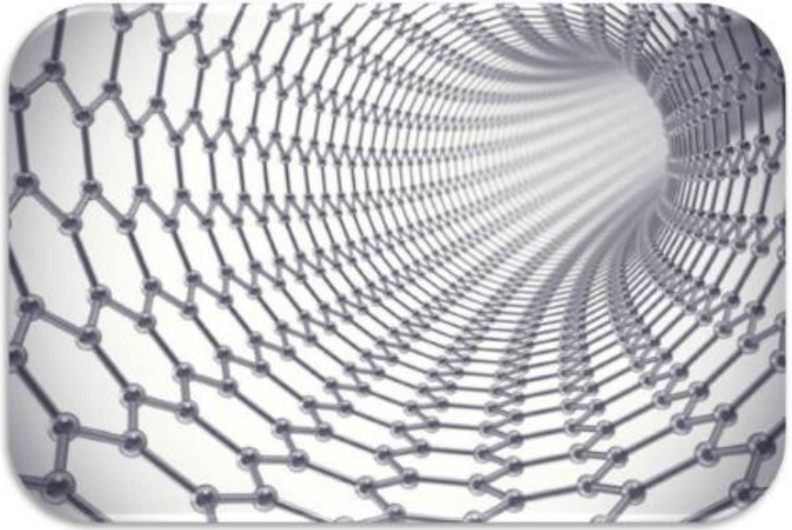
OTHER APPLICATIONS

- **Nanotechnology is improving the efficiency of fuel production from crude oil through better catalysis.** It is also enabling reduced fuel consumption in vehicles and power plants through higher-efficiency combustion and **decreased friction between moving parts.**
- Researchers are investigating carbon nanotube scrubbers and **membranes to separate carbon dioxide from power plant exhaust.**
- Researchers are **developing wires containing carbon nanotubes** that will have **much lower resistance** than the high-tension wires currently used in the electric grid, thus **reducing transmission power loss.**



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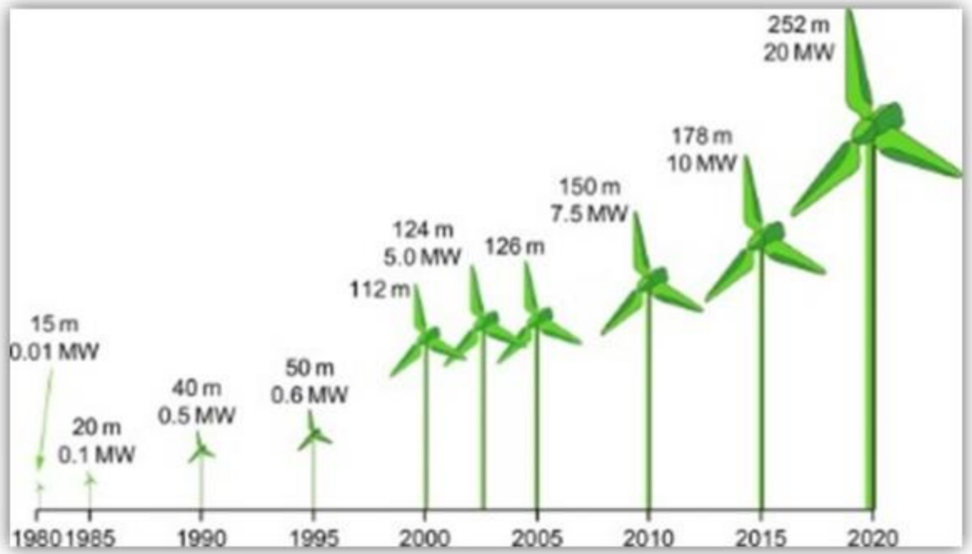
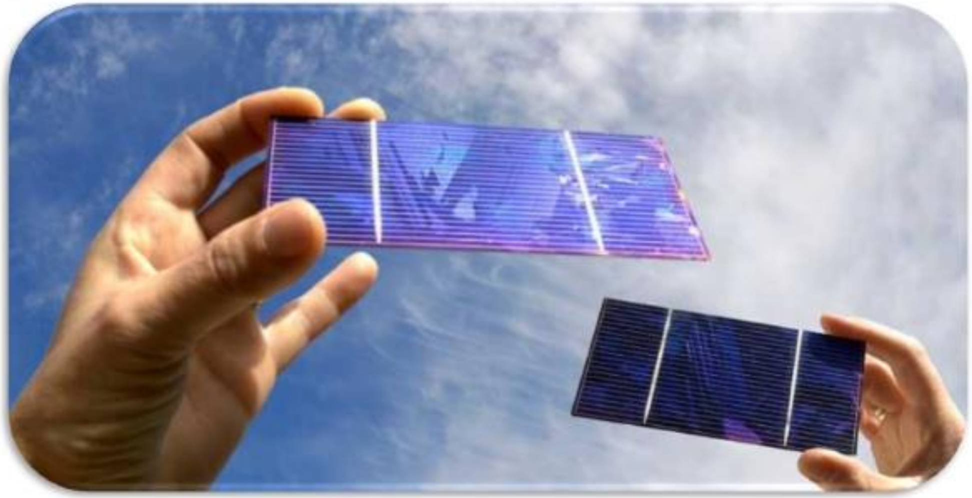
OTHER APPLICATIONS

- **Carbon nanotubes are being used to make windmill blades** that are longer, stronger, and lighter-weight than other blades **to increase the amount of electricity that windmills can generate.**
- Nanotechnology is already being used to develop many new kinds of batteries that are quicker-charging, **more efficient, lighter weight**, have a higher power density, and **hold electrical charge longer.**
- **Nanotechnology can be used in solar panels to convert sunlight to electricity** more efficiently, promising inexpensive solar power in the future. Nano-structured solar cells could be cheaper to manufacture and easier to install. They can also be made in **flexible rolls** rather than discrete panels.





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OTHER APPLICATIONS

- Newer research suggests that **future solar converters might even be "paintable."**
- Researchers are developing **thin-film solar electric panels that can be fitted onto computer cases and flexible piezoelectric nano-wires woven into clothing to generate usable energy** on the go from light, friction or body heat to power mobile electronic devices.
- Similarly, various nano-based options are being pursued to **convert waste heat in computers, automobiles, homes, power plants, etc., to usable electrical power.**
- **Storage bins are being produced with silver nanoparticles embedded in the plastic.** The silver nanoparticles **kill bacteria from any material** that was previously stored in the bins, minimizing health risks from harmful bacteria.

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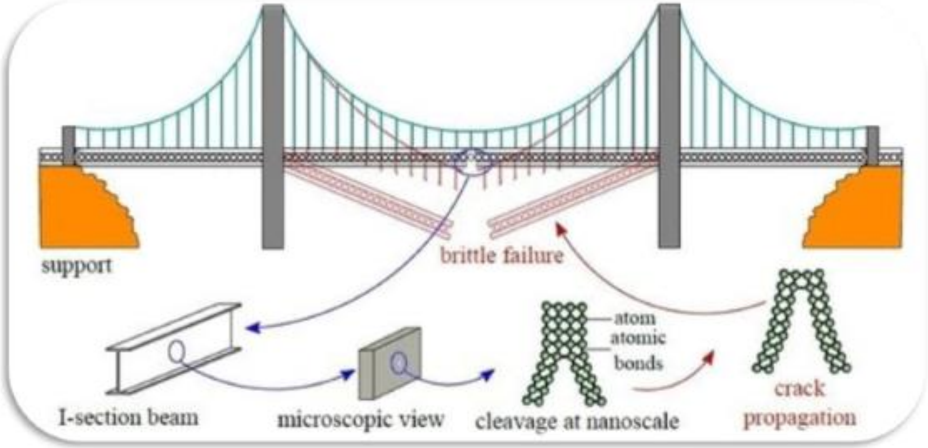
OTHER APPLICATIONS

- **Researchers are using silicate nanoparticles to provide a barrier to gases and moisture in plastic films used for packaging.** This could reduce the possibly of food spoiling or drying out.
- **Nanoscale sensors** may provide cost-effective, continuous monitoring of the **structural integrity of bridges, tunnels, rails, parking structures**, and pavements over time.
- "Game changing" benefits can be obtained from the use of nanotechnology-enabled lightweight, high-strength materials.



OTHER APPLICATIONS

- For example, it has been estimated that reducing the weight of a commercial jet aircraft by 20 percent could reduce its fuel consumption by as much as 15 percent.
- An analysis performed for NASA has indicated that use of nanomaterials with twice the strength of conventional composites would reduce the gross weight of a launch vehicle by as much as 63 percent. Not only could this save a significant amount of energy needed to launch spacecraft into orbit, but it would also enable the development of single stage launch vehicles, further reducing launch costs.

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